PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Acute subdural haematoma in the elderly- to operate or not to operate? A systematic review and meta-analysis of outcomes following surgery
AUTHORS	Manivannan, Susruta; Spencer, Robert; Marei, Omar; Mayo, Isaac; Elalfy, Omar; Martin, John; Zaben, Malik

VERSION 1 – REVIEW

REVIEWER	Callovini, Giorgi San Giovanni-Addolorata Hospital
REVIEW RETURNED	06-Apr-2021

GENERAL COMMENTS	The authors present a systematic review and meta-analysis regarding the outcome of patients aged 65 and above, operated on for acute subdural hematoma. They did a good job with the analysis and study desisgn; the conclusions are sound especially with regard to clinical practice and the manuscript is well written. Some aspects should be better investigated: 1) the role of decompressive craniectomy, even if the numbers are limited 2) I belive that the material can be used to answer the question if the adverse results in ASDH surgery can be predicted through the design of a predictive model. The work appears worthy of publication on BMJ Open.

REVIEWER	Halvorsrud, Kristoffer Queen Mary University of London
REVIEW RETURNED	16-Apr-2021

GENERAL COMMENTS	In general, this is a well-conducted and well-written systematic review. However, there are some points to be made (some more structural while others minor), which I hope can help the authors improve their manuscript. I have provided these points sequentially below. Throughout the manuscript, some abbreviations should be defined for readers by their full name the first time they are introduced: GOS, LTFU, GCS, MLS, CS, DC. When reporting confidence intervals throughout, the results of these are reported multiple places before indicating that these are

	95% confidence intervals (e.g. “32.73; 47.14%, 95% CI”). I would suggest reporting that these are 95% CIs before their results (e.g. “95% CI: 32.73; 47.14%”). In the abstract, it is mentioned that “Pooled estimate of percentage of poor outcomes …”. It would be useful for readers to know what more precisely is meant by ‘poor outcomes’ in this context. In the introduction it is mentioned that “With improvements in road safety measures, we are currently experiencing an increasing contribution of traumatic injuries in the elderly population towards the incidence of major intracranial injuries”. This sentence might come across to some, without further investigation, as presenting a somewhat inconsistent logic suggesting that improvements in road safety measures lead to an increased number of accidents? In the introduction it is also stated that “Importantly, mortality rates secondary to falls-related TBI were disproportionately higher in patients aged 75 years and above, at 54.08 per 100,000 patients.” But compared to what numbers for those under 75 years? In the subsection on literature search in the methods, for the sake of avoiding duplication and claiming originality of the present systematic review with more confidence, I wondered if the search also determined that no existing systematic reviews on this topic have already been published and whether also an additional search on systematic review protocol registers (e.g. PROSPERO) was conducted to determine that no ongoing systematic reviews are currently under way? Is there also a rationale why January 1990 in particular was chosen as the earliest date of searched publications? In the subsection on study selection in the methods, is there a rationale as to why patient age of 60 years or above was chosen as the cut-off point? In the subsection on statistical analysis in the methods, it is stated that “Given that I2 > 50% in all analyses…”. However, from forest plot B on ‘poor outcome at discharge’, it is indicated that I2 = 45%. Throughout the risk factors subsection of the results, some critiques might claim that the authors are susceptible to the risk of ‘vote counting’ – unless this is addressed/justified by the authors. For example, in the following part: “Statistically significant associations between GCS and GOS30,37,38 or mortality25,29,32-34 were demonstrated across multiple studies. Similarly, significant associations were also demonstrated between pupillary reactivity and GOS21,25,34,35 or mortality25,34. No significant relationship between GCS and GOS was demonstrated in four studies21,31,32,35, and between pupil reactivity and GOS/ mortality in four studies29,31,37,38.” Therefore. for clarity and to contextualise their methodological decisions for readers, I suggest that if not possible to combine studies (such as e.g. in comparative subgroup meta-analyses) that the authors provide a more explicit rationale as to why this

	was not feasible, as well as to highlight this as a limitation in their discussion section. By extension, the quality of the evidence is not systematically reported in the main text – which could have provided a better indication as to the confidence we might ascribe to significant versus non-significant results depending on their respective quality. Throughout the results section, the order of groups that are compared might also benefit from being reversed certain places. For example, when reference is made to comparison by number of co-morbidities, unless there is a good reason to keep in the same order I would suggest that it might be of more logical sense to mention groups with the greatest number of co-morbidities (e.g. “more than or equal to 5”) before those with less co-morbidities (e.g. “less than 5”). On page 11, line numbers 15-16, it is stated that “(ii) all eight patients with orthpaedic fractures had unfavourable outcomes”. It would be helpful for readers to know what is meant by ‘unfavourable outcomes’ in this context. On page 12, line numbers 7-11, it is stated that “Another study demonstrated a significantly smaller ASDH volume in the favourable outcome group when compared with the unfavourable outcome group (67ml vs 118ml; unpaired Student’s t test, p=0.05).” Please could the authors provide a reference for this study. The following sentence might confuse some readers (page 13, line numbers 40-47): “Only one study included patients above 60 years of age24, which was excluded from meta-analysis due to its date of publication. Therefore, our pooled estimated mortality and unfavourable outcome rates are generated from studies including patients aged 65 years and above”. However, some ‘patients above 60 years of age’ might also fall into the category of ‘patients aged 65 years and above’.

REVIEWER	Brogueira Rodrigues, Filipe Instituto de Medicina Molecular, Clinical Pharmacology Unit
REVIEW RETURNED	07-May-2021

GENERAL COMMENTS	The authors conducted a systematic review to evaluate the mortality and functional outcomes after the surgical management of acute subdural hematoma. Three databases were included between January 1990 and May 2020. All studies reporting the above stated outcome in patients with 60 or more years of age were included, irrespective of study design. A proportion meta-analysis was conducted to assess mortality and unfavourable outcome. I was invited to review the statistical and methodological aspects of this work and made no comments outside this scope. This seem a to be a well conducted systematic review asking a prognostic question. There are limitations to the methods (see below) that limit the conclusions. Among the most important are

	the exclusion of non-english reports and studies for which the authors were unable to access the evidence and the approach to investigate heterogeneity/effect modifies and confounders. It should also be noted that the nature of a prognostic question does not allow a direct assessment of the effect of an intervention and this could be clearer in the text. Below I provide some specific comments that the authors may want to consider: - in the abstract, when providing risks, it would help interpretation to provide study, events and sample numbers, together with a measure of statistical heterogeneity - pg5 ln38 - authors may want to clarify the rate unit (e.g. 100,000 person-year) - pg5 ln49 - PRISMA guidelines are reporting and not conduct guidelines; author may want to follow the updated PRISMA2020 statement - pg5 ln58 - authors may want to provide the full search strategy for each database - pg6 ln9 - it may be noted that excluding non-english reports limits internal and external validity - pg6 ln60 - authors do not report the extraction of study characteristics such as study design; authors should expand on how the STOBE statement was utilised to evaluate study quality - pg7 ln10 - authors do not specify which random-effect meta-analysis model was used; it could be explained why a transformation was needed - pg7 ln51- authors do not specify how subgroup analyses and meta-regressions were conducted, and from reading the results it seems that GLMs were used for such purpose. Authors should describe models specifications and how study weight was account for in such models bearing in mind this is an unusual method to explore association with other variables in a meta-analysis context. - pg7 ln51 - the authors justify the exclusion of studies before 2000 due to clinical and statistical aspects but do not provide the data supporting this decision. I suggest adding this information to the supplements. - pg8 ln27 - authors may want to clarify what average was used - it is not clear how study quality as was account for in the analyses and data do not seem to be provided in the manuscript - PRISMA flow chart - 10 studies weren't included as full text was not available. This seems an important limitation and should be noted in the description of the systematic review and discussed.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1

- **“[What is] the role of decompressive craniectomy, even if the numbers are limited?”**

In total, 6 studies distinguished between the number of patients that underwent decompressive craniectomies and craniotomies. Across these studies, 52 patients underwent decompressive craniectomies, but outcomes were only reported for 23 patients (McGinity et al; Monsivais et al; Bus et al). Two studies reported poor outcomes in in 2 out of 2, and 8 out of 10 patients (McGinity et al; Bus et al). One study reported mortality in 9 out of 11 patients (Monsivais et al). Given the small sample size and inconsistent reporting of outcomes, a separate subgroup analysis could not be performed. However, we have updated our Results to include this information (see Results, Risk Factors for Poor Outcome, Surgical Factors).

- **"I believe that the material can be used to answer the question if the adverse results in ASDH surgery can be predicted through the design of a predictive model."**

This is an excellent proposition and one that we had discussed prior to submission of our manuscript. Unfortunately, heterogeneity in reporting of risk factors prevented development of a predictive model. As an example, this is demonstrated by the variety of methods used to analyse neurological status in Supplementary Data 5. This should, however, be an avenue to explore in future retrospective or prospective studies provided a sufficient sample size.

Reviewer 2

- **"Throughout the manuscript, some abbreviations should be defined for readers by their full name the first time they are introduced: GOS, LTFU, GCS, MLS, CS, DC."**

We have updated our Methods section to expand on GCS as an abbreviation. Expansions for GOS and LTFU have already been provided in our Methods section, and for MLS and DL in the Introduction.

- **"When reporting confidence intervals throughout, the results of these are reported multiple places before indicating that these are 95% confidence intervals (e.g. "32.73; 47.14%, 95% CI"). I would suggest reporting that these are 95% CIs before their results (e.g. "95% CI: 32.73; 47.14%")."**

We have updated our manuscript with the suggested format.

    - **"In the abstract, it is mentioned that "Pooled estimate of percentage of poor outcomes …". It would be useful for readers to know what more precisely is meant by 'poor outcomes' in this context."**

The Methods section of our Abstract has been updated to include death/ severe disability as the definition of poor outcome.

    - **"In the introduction it is mentioned that "With improvements in road safety measures, we are currently experiencing an increasing contribution of traumatic injuries in the elderly population towards the incidence of major intracranial injuries". This sentence might come across to some, without further investigation, as presenting a somewhat**

inconsistent logic suggesting that improvements in road safety measures lead to an increased number of accidents?"

The first half of this sentence has been removed to ensure that this does not cause any confusion.

- **"In the introduction it is also stated that "Importantly, mortality rates secondary to falls-related TBI were disproportionately higher in patients aged 75 years and above, at 54.08 per 100,000 patients." But compared to what numbers for those under 75 years?"**

We have included numbers for adults aged less than 75 years (8 per 100,000 adults).

- **"In the subsection on literature search in the methods, for the sake of avoiding duplication and claiming originality of the present systematic review with more confidence, I wondered if the search also determined that no existing systematic reviews on this topic have already been published and whether also an additional search on systematic review protocol registers (e.g. PROSPERO) was conducted to determine that no ongoing systematic reviews are currently under way? Is there also a rationale why January 1990 in particular was chosen as the earliest date of searched publications?**"

We performed a literature search and a search of the PROSPERO database to ensure that similar systematic reviews had not been performed. Only one systematic review has been published which examines the prognosis of acute subdural haematoma in the elderly but this does not explore surgical mortality/ morbidity or risk factors for poor outcomes in surgical candidates specifically. January 1990 was chosen to try and offset the contribution of bias from potentially reduced access to healthcare/ less uniform use of modern surgical approaches to acute subdural haematoma evacuation at previous dates.

- **"In the subsection on study selection in the methods, is there a rationale as to why patient age of 60 years or above was chosen as the cut-off point?"**

Given that included studies would encompass various countries and timepoints over the last three decades, we wanted to achieve adequate coverage of the elderly population. We chose 60 years as the cut-off because this is often used to describe the elderly population by international bodies such as the United Nations and World Health Organisation.

- **"In the subsection on statistical analysis in the methods, it is stated that "Given that I2 > 50% in all analyses…". However, from forest plot B on 'poor outcome at discharge', it is indicated that I2 = 45%."**

We apologise for the error in our Methods section- this has been amended to $I^2 > 25\%$ to reflect moderate-large heterogeneity.

- **"Throughout the risk factors subsection of the results, some critiques might claim that the authors are susceptible to the risk of 'vote counting' – unless this is addressed/justified by the authors. For example, in the following part: "Statistically significant associations between GCS and GOS30,37,38 or mortality25,29,32-34 were demonstrated across multiple studies. Similarly, significant associations were also demonstrated between pupillary reactivity and GOS21,25,34,35 or mortality25,34. No**

significant relationship between GCS and GOS was demonstrated in four studies21,31,32,35, and between pupil reactivity and GOS/ mortality in four studies29,31,37,38." Therefore. for clarity and to contextualise their methodological decisions for readers, I suggest that if not possible to combine studies (such as e.g. in comparative subgroup meta-analyses) that the authors provide a more explicit rationale as to why this was not feasible, as well as to highlight this as a limitation in their discussion section. By extension, the quality of the evidence is not systematically reported in the main text – which could have provided a better indication as to the confidence we might ascribe to significant versus non-significant results depending on their respective quality."

In this systematic review, we aimed to examine risk factors for poor outcome in elderly patients who underwent surgical evacuation of ASDH. Whilst many studies examined risk factors, there was very little consistency in the method of reporting. For example, age was treated as a continuous variable in some studies and categorical in others. Baseline functional status was described by number of co-morbidities, pre-morbid independence, or pre-admission residence depending on the study. Neurological status was described with respect to pupillary reactivity or GCS with a wide range of categories across studies, as illustrated in Supplementary Tables 3 and 4. Therefore, comparative subgroup meta-analyses was not feasible. We have stated this explicitly within our Limitations section in our updated manuscript. Nonetheless, we felt that risk factors from individual studies should be discussed in a narrative fashion to discuss current evidence.

- **"Throughout the results section, the order of groups that are compared might also benefit from being reversed certain places. For example, when reference is made to comparison by number of co-morbidities, unless there is a good reason to keep in the same order I would suggest that it might be of more logical sense to mention groups with the greatest number of co-morbidities (e.g. "more than or equal to 5") before those with less co-morbidities (e.g. "less than 5")."**

We have switched the order of groups with respect to number of co-morbidities as suggested.

- **"On page 11, line numbers 15-16, it is stated that "(ii) all eight patients with orthopaedic fractures had unfavourable outcomes". It would be helpful for readers to know what is meant by 'unfavourable outcomes' in this context."**

We have replaced the term 'unfavourable' with 'poor' in order to maintain consistency within our manuscript, which reflects our previously defined dichotomous outcomes based on GOS score (Methods).

- **"On page 12, line numbers 7-11, it is stated that "Another study demonstrated a significantly smaller ASDH volume in the favourable outcome group when compared with the unfavourable outcome group (67ml vs 118ml; unpaired Student's t test, p=0.05)." Please could the authors provide a reference for this study."**

Our updated manuscript includes the corresponding citation.

- **"The following sentence might confuse some readers (page 13, line numbers 40-47): "Only one study included patients above 60 years of age24, which was excluded from**

meta-analysis due to its date of publication. Therefore, our pooled estimated mortality and unfavourable outcome rates are generated from studies including patients aged 65 years and above". However, some 'patients above 60 years of age' might also fall into the category of 'patients aged 65 years and above'."

We apologise for the source of confusion. We excluded this study from meta-analysis due to its date of publication (Methods, Statistical Analysis) as we only used studies published from January 2000 onwards. Therefore, although some patients in the study may fall into the target age category, this is not why they were excluded. We mentioned the fact that this was the only study including patients aged above 60 years because all other studies involved older ages as a cut-off (Table 1).

Reviewer 3

- **"In the abstract, when providing risks, it would help interpretation to provide study, events and sample numbers, together with a measure of statistical heterogeneity."**

We have updated our abstract to include the suggested information.

- **"pg5 ln38 - authors may want to clarify the rate unit (e.g. 100,000 person-year)"**

We assume that this point is in reference to Page 4/ Line 38- this has been amended as suggested.

- **"pg5 ln49 - PRISMA guidelines are reporting and not conduct guidelines; author may want to follow the updated PRISMA2020 statement"**

We apologise for the incorrect reference provided here; we have updated this citation.

- **"pg5 ln58 - authors may want to provide the full search strategy for each database"**

We have included full details of our search strategy as Supplementary Data 1.

- **pg6 ln9 - it may be noted that excluding non-english reports limits internal and external validity**

*This is a valid point and we have included this in our Limitations section (see **Discussion**).*

- **"pg6 ln60 - authors do not report the extraction of study characteristics such as study design; authors should expand on how the STOBE statement was utilised to evaluate study quality"**

We have now included an assessment of risk of bias across included studies using ROBINS-I tool, which we have adapted for use in cohort studies (see Table 1 and Supplementary Data 7 for full breakdown). We have also detailed eligibility for inclusion in meta-analysis based on ROBINS-I tool assessment and STROBE checklist criteria ('Studies with STROBE score <13 or ROBINS-I score of critical risk of bias were deemed ineligible for inclusion in meta-analysis").

- **"pg7 ln10 - authors do not specify which random-effect meta-analysis model was used; it could be explained why a transformation was needed."**

DerSimonian Laird random-effects model was used to account for between-study variation and to enhance the generalisability of conclusions. The double arcsine square root transformation was used due to the relatively small sample size in some studies and high proportion of poor outcomes involved.

- **"pg7 ln51- authors do not specify how subgroup analyses and meta-regressions were conducted, and from reading the results it seems that GLMs were used for such purpose. Authors should describe models specifications and how study weight was account for in such models bearing in mind this is an unusual method to explore association with other variables in a meta-analysis context. pg7 ln51 - the authors justify the exclusion of studies before 2000 due to clinical and statistical aspects but do not provide the data supporting this decision. I suggest adding this information to the supplements."**

We have included extensive details of our subgroup analysis and rationale for excluding pre-2000 studies (Supplementary Data 1). Please note that our decision to exclude these studies from meta-analysis was based primarily upon clinical reasoning and supplemented by statistical analysis.

- **"pg8 ln27 - authors may want to clarify what average was used"**

We have specified the type of average used throughout our updated manuscript.

- **"it is not clear how study quality as was account for in the analyses and data do not seem to be provided in the manuscript"**

We have now included an assessment of risk of bias across included studies using ROBINS-I tool, which we have adapted for use in cohort studies (see Table 1 and Supplementary Data 7 for full breakdown). We have also detailed eligibility for inclusion in meta-analysis based on ROBINS-I tool assessment and STROBE checklist criteria ('Studies with STROBE score <13 or ROBINS-I score of critical risk of bias were deemed ineligible for inclusion in meta-analysis").

- **"PRISMA flow chart - 10 studies weren't included as full text was not available. This seems an important limitation and should be noted in the description of the systematic review and discussed."**

*This is a valid point and we have included this in our Limitations section (see **Discussion**).*

VERSION 2 – REVIEW

REVIEWER	Halvorsrud, Kristoffer Queen Mary University of London
REVIEW RETURNED	08-Jul-2021

GENERAL COMMENTS	Based on the authors' overall consideration of my comments, I can recommend for publication.